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ERROR

❑ **Error** in communication is a condition when receiver's information doesn't match with sender's information. There are many reasons such as noise, cross-talk etc., which may help data to get corrupted during transmission.

❑ **Types of error:**

1. Single bit error: When only one bit of transmitted data is altered. The data can be corrupted in any position during transmission.



TYPES OF ERROR

2. Multiple bit error: When more than one bit of data is corrupted during transmission of information.



3. Burst error: A burst error occurs when two or more consecutive bits in a data unit are corrupted during transmission.



ERROR DETECTION

❑ First step in error control is to detect the error occurred during transmission. Here are most common error detection methods:

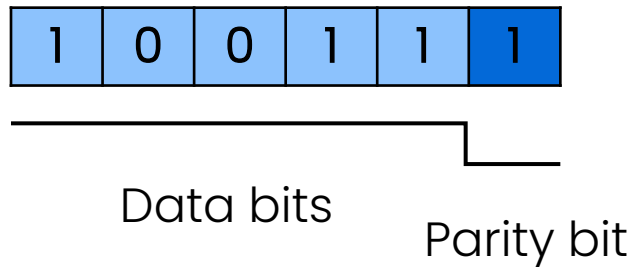
1. Simple parity check
2. Two-dimensional parity check
3. Checksum
4. Cyclic Redundancy Check (CRC)

❑ To control error in transmission we must correct the corrupted bit when detected, this is error correction process. There are two types of error correction in digital communication:

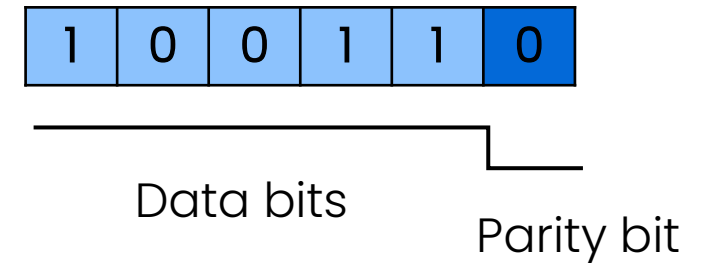
1. Forward error correction
2. Backward error correction

SIMPLE PARITY CHECK

- ❑ In **simple parity check** technique an extra bit, known as parity bit, is added to data unit before transmission. This parity bit helps the receiver determine whether the transmitted data has been corrupted. There are two types of parity:
 - i. **Even parity:** The parity bit is set so that the total number of 1s in the data (including the parity bit) is even.
 - ii. **Odd parity:** The parity bit is set so that the total number of 1s in the data (including the parity bit) is odd.



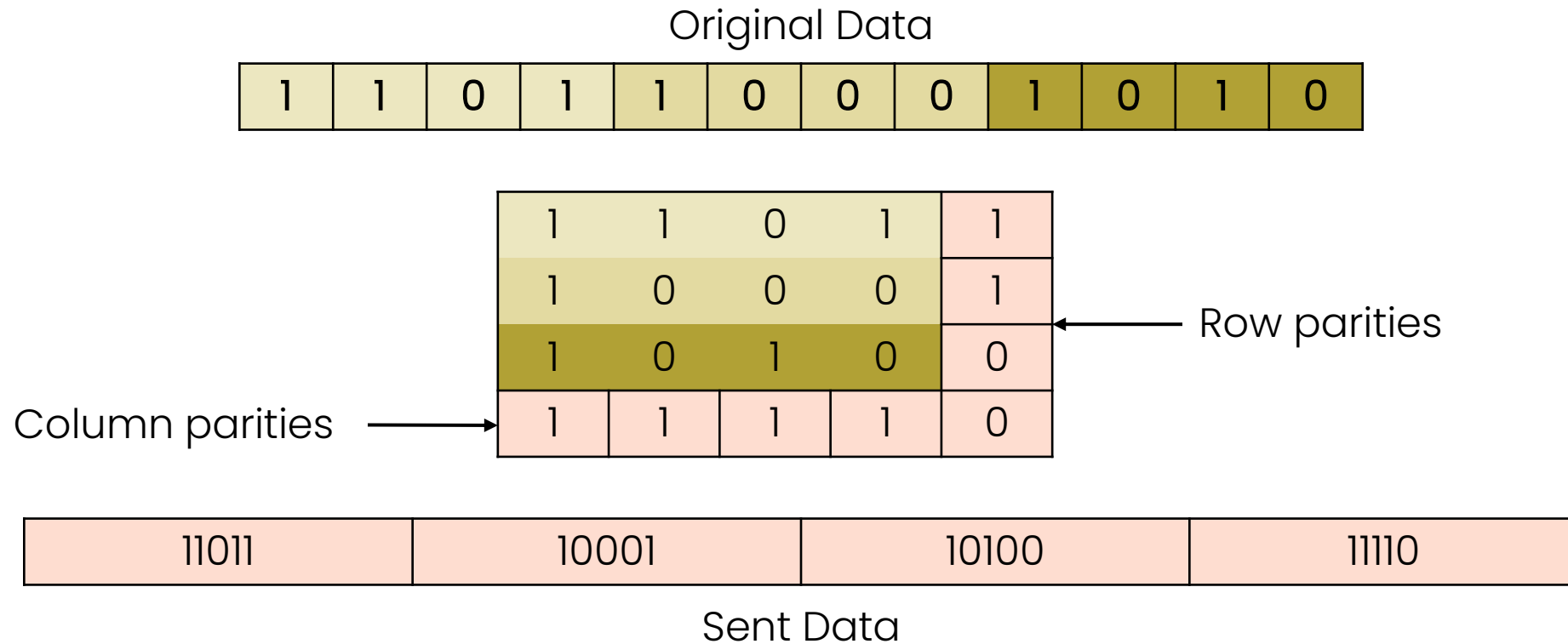
Even Parity



Odd Parity

TWO-DIMENSIONAL PARITY CHECK

- ❑ In **two-dimensional parity check**, parity bits are calculated for each row and column, similar to a simple parity check. These row and column parity bits are transmitted along with the data. At the receiver, parity bits are recalculated and compared with the received parity bits to detect errors.



CHECKSUM

- ❑ In this method, the sender divides the data into fixed-size segments and computes a total sum of them and does 1's complement. The computed checksum is transmitted along with the data. At the receiver, the same computation is performed to verify data was not corrupted.

Original Data

10011001	11100010	00100100	10000100
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At sender's end, original data is segmented in 4 parts and summed together. In the end we 1's complement the sum to get our checksum which is sent with the data for receiver to detect if any error occurs during transmission

Sum →
Checksum →

	1	0	0	1	1	0	0	1
	1	1	1	0	0	0	1	0
1	0	1	1	1	1	0	1	1
								1
	0	1	1	1	1	1	0	0
	0	0	1	0	0	1	0	0
	1	0	1	0	0	0	0	0
	1	0	0	0	0	1	0	0
1	0	0	1	0	0	1	0	0
								1
	0	0	1	0	0	1	0	1
	1	1	0	1	1	0	1	0

CHECKSUM

Data at receiver end

10011001	11100010	00100100	10000100	11011010
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Data is accepted at receiver end if the complemented sum equals to 0, which indicates there is no error.

Checksum →

Complemented →

	1	0	0	1	1	0	0	1
	1	1	1	0	0	0	1	0
1	0	1	1	1	1	0	1	1
								1
	0	1	1	1	1	1	0	0
	0	0	1	0	0	1	0	0
	1	0	1	0	0	0	0	0
	1	0	0	0	0	1	0	0
1	0	0	1	0	0	1	0	0
								1
	0	0	1	0	0	1	0	1
	1	1	0	1	1	0	1	0
	1	1	1	1	1	1	1	1
	0	0	0	0	0	0	0	0

CYCLIC REDUNDANCY CHECK

- ❖ Let a message 11100000 that is to be sent over a medium and the divisor is given polynomial x^3+1 . Find error through CRC method.

At sender's end,

Original Data

11100

Divisor

$$x^3+1 = 1*x^3+0*x^2+0*x^1+1*x^0=1001$$

Extra bit= no. of bits in divisor – 1
XOR function of data & divisor

Divisor				Data					Extra bit		
1	0	0	1	1	1	1	0	0	0	0	0
				1	0	0	1				
				0	1	1	1	0			
					1	0	0	1			
					0	1	1	1	0		
						1	0	0	1		
						0	1	1	1	0	
							1	0	0	1	
							0	1	1	1	0
								1	0	0	1
								0	1	1	1

Remainder

CYCLIC REDUNDANCY CHECK

At receiver's end,

1	1	1	0	0	0	0	0
					1	1	1
1	1	1	0	0	1	1	1

Transmitted data

No error, data
accepted

1	0	0	1	1	1	1	0	0	1	1	1
				1	0	0	1				
				0	1	1	1	0			
					1	0	0	1			
					0	1	1	1	1		
						1	0	0	1		
						0	1	1	0	1	
							1	0	0	1	
							0	1	0	0	1
								1	0	0	1
									0	0	0
										0	0

HAMMING CODE

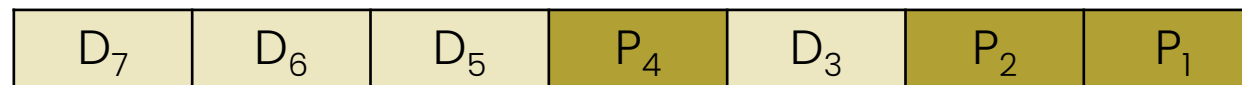
- ❑ **Hamming Code** is a set of error correction codes that can be used to detect and correct errors that occur during data transmission. It uses the following formula to calculate redundant or parity bits.

$$2^P \geq m + P + 1$$

Where, m= data bit, P= parity bit.

Parity can be odd or even parity.

- ❑ Let's assume a 4-bit data. Then parity/redundant bit will be 3 as, $2^3 \geq 4 + 3 + 1$
- ❑ Let data bit be D and parity bit P.
- ❑ Position of parity bits: $P_1=2^0=1^{\text{st}}$, $P_2=2^1=2^{\text{nd}}$, $P_3=2^2=4^{\text{th}}$ bit.
- ❑ So, transmitted data:



- ❑ When data bit is corrupted during transmission, parity bit value changes and at receiver end it is detected.

HAMMING CODE GENERATION

Data bit= 1011

So, total parity bit will be 3 (P_1 , P_2 & P_4)

1	0	1	P_4	1	P_2	P_1
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$P_1 \rightarrow D_3 D_5 D_7 \rightarrow 1 1 1 \rightarrow P_1=1$ (Checks 1 bit and skips 1 bit)

$P_2 \rightarrow D_3 D_6 D_7 \rightarrow 1 0 1 \rightarrow P_2=0$ (Checks 2 bits and skips next 2 bits)

$P_4 \rightarrow D_5 D_6 D_7 \rightarrow 1 0 1 \rightarrow P_4=0$ (Checks 4 bits and skips next 4 bits)

1	0	1	0	1	0	1
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Transmitted data

HAMMING CODE ERROR DETECTION & CORRECTION

Let's assume sent data was 1011, but data received was 1010.

1	0	1	0	0	0	1
D ₇	D ₆	D ₅	P ₄	D ₃	P ₂	P ₁

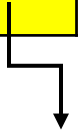
$P_1=1 \rightarrow D_3 D_5 D_7 \rightarrow 0 1 1 \rightarrow P_1=1$ [For wrong parity we put value 1.]

$P_2=0 \rightarrow D_3 D_6 D_7 \rightarrow 0 0 1 \rightarrow P_2=1$

$P_4=0 \rightarrow D_5 D_6 D_7 \rightarrow 1 0 1 \rightarrow P_4=0$ [For right parity we put value 0.]

So, error is in $(P_4 P_2 P_1)^{\text{th}}$ bit = $0 1 1 = 3^{\text{rd}}$ bit

1	0	1	0	1	0	1
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Bit corrected

PRACTICE PROBLEM

❑ **Cyclic redundancy check:** sent data= 100100 and key= 1101; after error at receiver data= 100000.
Find remainder and check error at receiver end.
Answer: remainder= 001, error at receiver end= 01

❑ **Hamming code:** received data= 1011011; check for error and correct it.
Answer: 3rd bit error.

❑ **Hamming code:** For given data find the parity bits. Data= 10011
Answer: 1111

❑ **Checksum:** Data= 11001 10000 10101 11011, show how checksum works for given data.

Check similar problems.



THANK YOU

